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An intelligent system to help deaf students learn Arabic Sign Language

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ABSTRACT

This paper proposes a new system to translate an Arabic Sign Language (ArSL). The system consists of two sub-systems: the first, Speech to ArSL translation Subsystem. This sub-system is mainly based on the speech recognition engine. The second is ArSL to speech translation subsystem to translate the images of signs into speech, mainly based on GTTS library. This system will be easily used by both groups (the hearing disabled and normal persons) who want to learn ArSL and will support more communication between them. Among the advantages of this system is that it is expandable by adding new signs to the database and can be accessed online at: https://sr.gravita-demo.com/. The proposed system was evaluated using several methods. The findings show that the system can translate ArSL with a recognition rate of 99%. An experimental approach was used to evaluate the effectiveness of the proposed system. Results show that the proposed system was effective and the ability of Arab deaf students to learn improved significantly. In addition, the system's performance evaluation questionnaire revealed that system users preferred this approach to ArSL learning and teaching and acquired new concepts, which predicts a promising future for this system in academic environments.

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KEYWORDS
Arabic Sign Language; sign language recognition; speech recognition engine; Google text to speech; deaf

1. Introduction

Based on WHO's statistics, more than 5% of the world's population (466 million approximately) are hearing disabled (432 Million adults and 34 Million children). More than 900 million people will be suffering from hearing loss by 2050. Moreover, around 10 million people in the Middle East are Deaf. According to the World Federation of the Deaf, there were more than 700,000 persons in the Arab world who were deaf and hearing impaired in 2008 (Alzohairi et al., 2018; Neiva & Zanchettin, 2018; Salema et al., 2019; WHO, 2020). Those suffering from hearing and/or speaking disorders cannot communicate or interact orally with others, especially when they try to get interact in educational, social and work environments. Moreover, they often have difficulties in learning how to read and write texts (Alzohairi et al., 2018; El-Gayyar et al., 2016; Shukor et al., 2015). Hence, usually they are excluded from society and are deprived from their rights to equal educational and professional opportunities. This being the case, these people are in a dire need for a language to communicate with people around them. Here, comes the role of sign languages as an effective alternative to express their ideas and thoughts, to fill the gap communication between them and others (Alzohairi et al., 2018; Suharjito et al., 2017). The gesture is one of the nonverbal and non-vocal communication forms in which different body parts are used to express certain sentences and messages

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(Bairagi, 2014). Sign language is a set of hand gestures and movements, positions and facial expressions that correspond to letters and words in natural languages. So, the sign language has more than one form because of its dependence on natural languages (Al-Dosri et al., 2014; El-Bendary et al., 2011; Kumar et al., 2017). In the twentieth century, Sign Language (SL) acquired increasing interest and global recognition by several linguists and computer scientists because it is regarded as the Native language of the deaf who can use it to voice their needs and form concepts (Ahmed et al., 2018; Almasoud & Al-Khalif, 2011). Like spoken languages, SL is not universally unified as each country has its own SL that is used by its hearing disabled citizens. So, there are Chinese sign language (CSL), Arabic Sign Language (ArSL), French sign language (LSF), … etc. (Brour & Benabbou, 2019; Cheok, 2019). In the Arab world, ArSL is the official sign language for people suffering from hearing and speech impairment. ArSL was formally launched in 2001 by the Arab Federation of the Deaf (AFD). In spite of being one of the most spoken languages globally, ArSL is still under evolution (Abdel-Fattah, 2005; Alzohairi et al., 2018). Studies conducted by the National Association of Deaf indicate the need for using Sign Language to educate those suffering from hearing impairment (Amrutha et al., 2016). Many schools were opened specifically to teach sign language to the Deaf in several countries. However, these schools still need to be more developed, especially in our Arab societies. Although ArSL is the most used method to teach the deaf in the Arab world, opportunities for qualifying and training SL translators/teachers are few, which adversely affect the ability of deaf student to learn SL and create a strong need for the development of modern methods and technologies to teach SL to deaf students. On the other part, SL teaching lessons consumes a lot of money and time and usually are provided in formal one-to-many sessions with a teacher who train learners/trainees how to perform the signs (Foong & Low, 2009; Ritchings et al., 2012). Hence, it is highly important to develop tools and techniques to improve and facilitate SL learning to include the hearing disabled in educational environments (Pontes et al., 2018). With the current advancements in information technology and communication, it has become essential to develop several assistive systems to support the deaf communities globally (El-Gayyar et al., 2016). Recently, there has been an increased interest in developing automatic Sign Language recognition systems (SLRS) to further the communication between normal and hearing impaired or deaf people (Yang et al., 2016). SLRS is one of the application areas of human–computer interaction (HCI). The main goal of SLRS is to recognize signs of hearing-impaired people and convert them to text or speech and vice versa (Al-Rousan et al., 2009; Ibrahim et al., 2018; Kumar et al., 2017; Tubaiz et al., 2015) which is the field of our current study. Several studies, including El-Gayyar et al. (2016), indicate that assistive technologies targeting Deaf communities in the Arab world are very few and limited due to the complexity of both Arabic and ArSL (El-Gayyar et al., 2016). ArSL studies admit the need for developing technologies that can be used to meet the requirements of the hearing impaired or the deaf in the Arab world. They also aim at creating environments that can include those suffering from hearing loss. This field of research is encouraged and supported in the Arab world by several social bodies in health, educational, transport and other sectors. However, few individual attempts were done to develop computer ArSL learning systems (Almasre & Al-Nuaim, 2020). In view of this, the main motive behind the development of our proposed system is to try to supplement research efforts in the field of ArSL recognition and attract the attention of the researchers of this field to serve the interests of the Deaf. This paper presents an interactive web-based system that can support teaching SL in classrooms and enable users to use SL without supervision at their own pace at home. The system performs a two-way translation between SL and spoken language. The system is mainly designed to help students suffering from hearing impairment to learn ArSL, improve their interaction with the hearing world and acquire new concepts. It also plays an important role in assisting SL teachers. The rest of this paper is organized as follows: we outline related work in Section 2, describe the proposed system in Section 3, show experimental work in Section 4, describe system evaluation in Section 5, follow this by an outline of our findings in Section 6 and finally present the conclusion and future work.

2. Related work

Sign language recognition has been an active area research and development in recent years. Many systems developed for sign language recognition targeting different sign languages Table 1 provides an overview of these studies. However, few of these systems were developed to recognize ArSL, mainly due to a lack of resources or the complexity of the Arabic language itself Table 2 provides an overview of these studies.

3. Description of the proposed system

This paper proposes an ArSL translation system. It is an educational application developed to support the deaf and the dumb community to facilitate their communication with normal people. It also helps learn and teach ArSL in academic environments. This system is used to convert user's speech into signs/text or convert the signs of the deaf user to speech/text, which makes it a two-way system and makes special communities closer to normal people. In general, this system enables users to choose one of two different systems: Speech to ArSL or ArSL to speech. This is shown in Figure 1, which presents an overview of the architecture of the proposed system.

In this section, we will describe and discuss the details of the proposed system and the used database. From Figure 1, it is clear that the system consists of two sub-systems, each of which will be discussed in detail in Sections 3.1 and 3.2 below.

Table 1. Related work on different SL recognition systems.

Reference	System, method/algorithm	Corpus	Accuracy
El-Darymli et al. (2006)	Interpreting Speech to Sign Language Interpreter System (SSLIS)	Translating spoken English to SL	The system converted speech to text in batch and live modes. It demonstrated Speech Recognition process, calculated WER automatically converted text to computer-generated voice conversion
Foong and Low (2009)	Translating English voice to sign language system with Speech and Image processing technique Consists of a sound recording part (with a sound/voice training algorithm), digital signal processing part (MFCC – Mel Frequency Cepstral Coefficients algorithm counterparts), and vector quantization part	Natural Language to SL translation	Accuracy depends on system training Currently, the system achieved an accuracy of 80.3%
Rajam and Ganesan (2011)	Used feature point extraction method to preprocess static images	Human computer interaction Tamil Sign Language Recognition	The system can recognize images with 98.125% accuracy if trained on 320 images and tested with 160 images
Kumar (2018)	The system used MATLAB to recognize sign twenty-six hand gestures of Indian SL The system used four modules: pre-processing and hand segmentation, feature extraction, sign recognition and sign to text It used Linear Discriminant Analysis (LDA) algorithm to recognize gestures	Sign language recognition	The system was successful in recognizing the target gestures using LDA algorithms
Wadhawan and Kumar (2020)	Used convolutional neural networks (CNN) based on deep learning Four main phases: data acquisition, image pre-processing, training and testing of the CNN classifier	Indian Sign Language (ISL) recognition	Achieved highest training accuracy (99.72%) and colored and grayscale images accuracy (99.90%)

Table 2. Related work on ArSL recognition systems.

Reference	System, method/algorithm	Corpus	Accuracy
El-Gayyar et al. (2016)	Cloud computing and a set of third-party services to translate Arabic speech and extract Arabic text from different sources into ArSL System uses text processing service to enhance Arabic text to achieve a better translation process	Translating Arabic speech and text into Arabic Sign Language	Passed the performance test (3.7–5.5 s waiting time) and the usability test (SUS = 79.8)
Ibrahim et al. (2018)	Automatic visual SLRS to translate the signs of isolated Arabic words into text Four main phases: hand segmentation, tracking, feature extraction and classification	Sign language recognition systems	Recognition rate of 97% and low misclassification rate for similar gestures A minimum TER of 0.08%. Strength in various cases of occlusion
Deriche et al. (2019)	a Bayesian approach with a Gaussian Mixture Model (GMM) and a simple Linear Discriminant Analysis (LDA) approach Introduced Dempster-Shafer (DS) theory of evidence which is a fusion technique based on evidence	Arabic Sign Language Recognition	92% accuracy in recognizing Arabic Sign Language System performance higher than latest glove-based systems and single-sensor based methods
Almasre and Al-Nuaim (2020)	A dynamic prototype model (DPM) using Kinect as a sensor to recognize certain Arabic Sign Language gestured dynamic words Eleven predictive models of three algorithms (SVM, RF, KNN)	Using sensors and natural user interfaces to enable and facilitate Arabic Sign Language recognition and interpretation	SVM models achieved highest accuracy rates in recognizing dynamic words gestured Linear kernel and cost parameter ¼ 0.035

3.1 Speech to ArSL subsystem

Speech to ArSL subsystem was developed to translate and convert natural speech into signs understandable by the deaf. The system works in several phases, as shown in Figure 1.

3.1.1 Speech input

At first, the user's Arabic speech is input through the microphone. Then, this speech is sent to the speech recognition engine to be processed.

3.1.2 Speech recognition engine

The Speech to ArSL Subsystem mainly depends on a software component known as the speech recognition engine that has a somewhat complex job, that is, to receive speech input, recognize it and translate it into a recognized text understandable by the application. Speech recognition techniques can be divided into two categories: speaker dependent and speaker independent. In speaker-dependent systems, models are created for a particular speaker and subsequently are not intended to work for different speakers. Speaker-independent systems need to be more generic and consequently require more complex design. Speech recognition systems tend to be much more complicated because of the large variation of pronunciations, even from the same speaker, depending on the context and mood of the speaker (Kim et al., 2019). In this work, the Speech to ArSL Subsystem uses speaker-dependent and independent speech recognition system using Python-based Arabic speech recognition library.

3.1.3 Text tokenization

Tokenization is a process in which a text stream is broken up into words, phrases, symbols, or other meaningful elements that are called tokens (Vijayarani & Janani, 2016). in this phase, the recognized text in the previous step is split into tokens consisting of words using the method split.

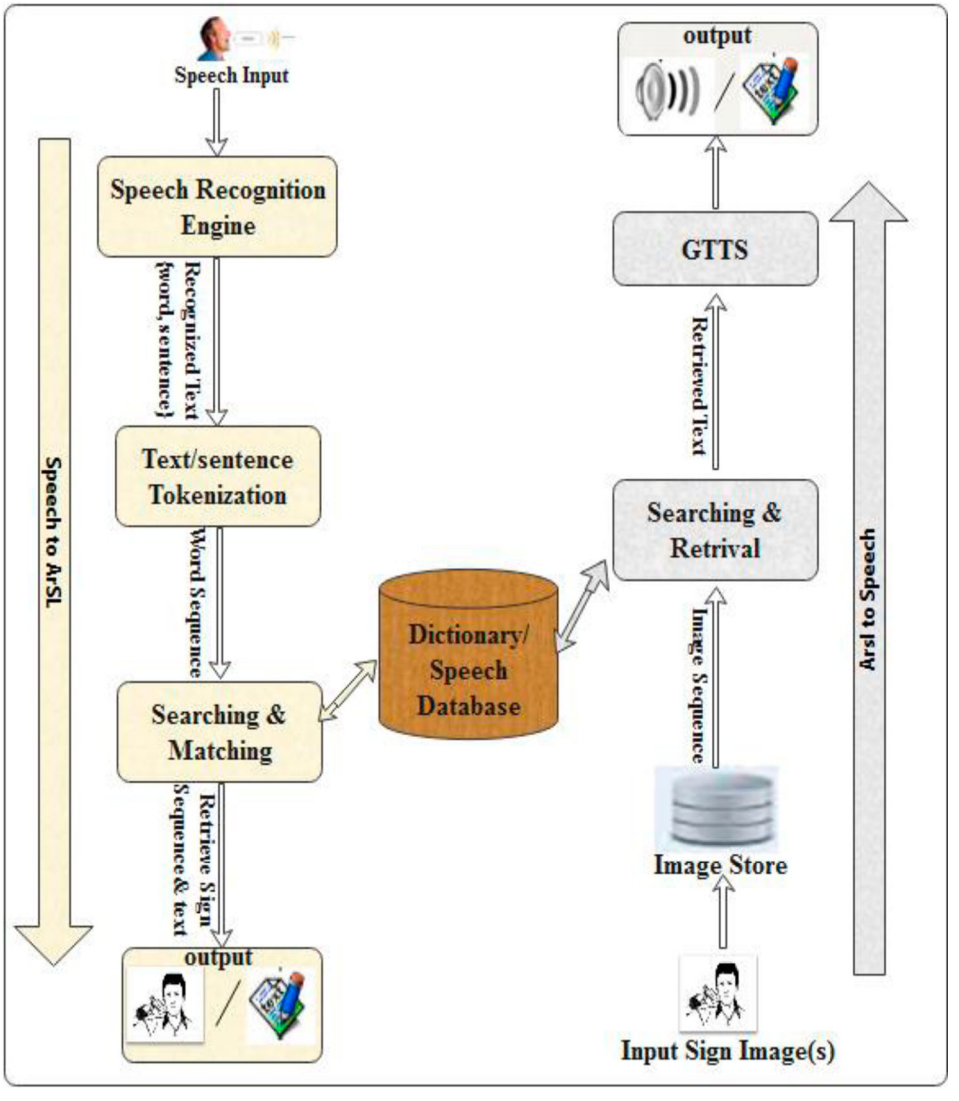


Figure 1. Architecture of the proposed system.

3.1.4 Searching & matching

The database is an important part of the entire application development. In this phase, the database/dictionary of the proposed system was built. It is noteworthy that a unified ArSL dictionary was published in two versions in 2008. However, due to the lack of a serious interest in ArSL recognition, there are no common databases available to researchers in this area. So, we built our own Sqlite database named "Speech Database" using SQLparse library, which is a database development library in Python. The Speech Database consists of two tables: Sdata, which contains the list of stored signs and their corresponding words, and SImage, which contains the images that will be selected to be converted into speech in the ArSL – To-Speech subsystem, which will be described in detail in Section 3.2 below. The Speech Database contains a list of about 500 signs for the following sections: letters, numbers, education and learning, school and its tools, animals, food and its tools, family, friends and people, money and quantities, sports and hobbies, time and seasons, travel, holidays and directions, home, clothes and colors, jobs, health and emergency, synonyms

and antitheses, different things. These are the most used signs in communication between the deaf and hearing impaired and normal people in academic environments and life activities. These sections were used based on a discussion with students suffering from hearing/ speech disabled and their teachers. The speech database is characterized by being easily expandable through a web-based interface that allows the admin to add new signs to the database and delete other signs from it. Figure 2 shows the administration panel of the database, which allows controlling the database.

After building the Speech Database, the system searches in the speech database for the word/ sequence of words obtained in the previous phase. When a match with the words pre-stored in the database is found, the equivalent sign/sequence of signs are retrieved and showed to the deaf person to understand the speech of the normal person. The system uses still images with no motion to express the recognized speech, so, it is more suitable for educational purposes rather than for translation.

3.1.5 Output

As output, the system displays the image/images of the sign equivalent to the speech input, with its textual value displayed below the image. This system provides the output in one of two cases: The first case, an image of a sign equivalent to a single spoken word. In the second case, the system presents a sequence of sign images equivalent to a certain sentence/sequence of spoken words. Figure 3 shows the pseudo code of the system developed for translating speech into sign image and text.

3.2 ArSL to speech translation subsystem

ArSL to speech translation subsystem was developed to translate sign images into natural speech understandable by normal people. This system consists of several phases, as shown in Figure 1.

3.2.1 Input sign image(s)

The input of this system is a sign image/sequence of sign images to be converted into natural speech that can be heard on a speaker.

3.2.2 Image store

After selecting the sign image/image sequence as inputs for the system, these images are stored in the "image store" (SImage table in the Speech Database), which is a temporary memory where recently identified images are stored to perform search and retrieval process, as will be shown in the following step, to obtain the right outputs. Note that all the images stored in the SImage table will be completely deleted once the required output is obtained by the system, and SImage will become ready for other new entries.

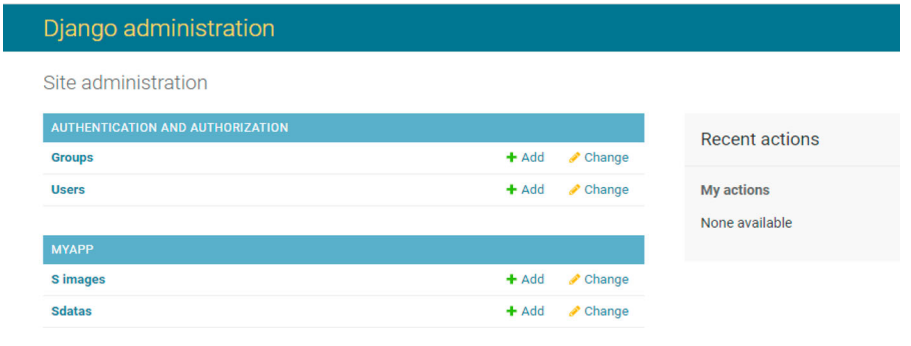


Figure 2. GUI of Django administration database panel.

Pseudo cod for translating Speech to ArSL

```
//IN:
 Speech Signals (word, sentence) SS
//OUT:
Sign Image/ images SI & Recognized Text
Begin
        //Construct DB that contain (Sign Data table, Sign Image table)
        //imports  Essential Libraries (Speech Recognition,  Django)
        //Construct HTML page "speech to text .html"
        Input SS by mic
        If SS= empty then
        Display the necessary error message
        End If
        else
        Save SS as 'audio' file
        Convert SS to Text using Speech Recognition Library
        Set the speaking language to Arabic (AR)
        Save the Recognized Text {Word (W) or Sentence (S)} in str_variable
        If str_vaiable = W then
         Search in DB about W
              If W is found in DB then
         Retrieve the equivalent text for (W)
         Retrieve the equivalent sign image for (W)

      Send& Display equivalent text & equivalent sign image to HTML page "speech_to_text .html"
          Else
              Display the necessary error message
       End If
        If str_variable = Sentence then
        Segment str_variable into tokens
       Words Ws() = str_variables. Split(" ")
      End If
      For each  word IN Ws
        IF  word is not found Goto 10
        Else
              Retrieve the equivalent text for  word
              Set Phrase_varaible =word
              Phrase_varaible += word
              Retrieve the equivalent sign image (SI) for this word
              Set Signs_variable =SI
             Signs_variable + = SI
        End if
    Next
        Send& Display Phrase_varaible (word sequence/sentence) & Signs_variable (sing image sequence) in
HTML page "speech_to_text.html"
         10:   Display necessary error message "this word doesn't exist in DB".
```

Figure 3. Pseudo code for translating speech to ArSL.

3.2.3 Searching & retrieval

The Speech-to-ArSL system also depends mainly on the same Speech Database mentioned above. In this phase, the system searches inside the "Sdata" table for the sign image/image sequence stored in the "image store" one after the other. If a match is found, the word/sequence of words corresponding these images are retrieved. In this way, the input sign image is converted into text, which is then sent to the Google text To speech (GTTS) and Google Text to Speech Token (GTTS-token) libraries. The basic idea upon which the search process is based is that the image is stored in the database under the same name of the image searched for. For example, if the word to be searched for "كتاب", its corresponding image name will be "كتاب" hence, the search is done using the image name.

3.2.4 GTTS

In this phase, the text (word/sequence of words) received from the previous phase will be translated and converted into speech using GTTS library. This library has been used to achieve the highest

accuracy and efficiency by converting the text into speech so that a normal person can hear the speech corresponding to these signs through the microphone. Along with GTTS library, GTTS-token library was used to ensure the efficiency of speech by dividing the input text into small parts each of which consists of one letter so that the output speech looks as closer as possible to the natural and continuous speech of a natural person.

3.2.5 Output

The output of the system is the equivalent speech of the input sign image/sequence of images with a display of the textual value of this speech input. This system provides output in one of two cases: the first case, a specific spoken word matching a single input sign image is output. in the second case, a sequence of words/spoken sentence is output matching the input sign images. Figure 4 shows the pseudo code of the system developed for converting and translating the sign image into speech and text.

Figure 5 below shows a part of the code used to translate speech into text using Python language.

4. Experimental work

The proposed system was developed using Python 3.6, HTML and JavaScript languages, as well as many built-in Python libraries, most importantly Speech Recognition and GTTS. The proposed system was designed to be a useful tool for the deaf community and people interacting with them, especially inside educational institutions. Accordingly, there is a need for an understandable and easy-to-use interface to provide simplicity and effective use of the system. We visited heard impaired school in Al Mansoura City, which is a school for the education of Deaf. According to special program teachers in the school, most deaf students have an intermediate understanding of Arabic. So, we designed the windows in Arabic. This made the system an instructional tool for ArSL learning. Figure 6 shows the homepage of the proposed web application. It has two options:"الصوت الي اشارة" and "الاشارة الي صوت".

When the "الصوت الي إشارة" link is clicked, an interface of the first subsystem appears, as shown in Figure 7. This subsystem translates speech into ArSL and text.

As shown in Figure 7, we click the "تحدث" button, and then we start speaking. For instance, when we say "أنا احب المدرسة", this spoken sentence will be translated into a string of corresponding sign

Pseudo code for translating ArSL to speech

```
//IN
Sign Image/ Images SI
//OUT
Speech Signal (Word, Sentence) SS  & Recognized Text
BEGIN
        //imports  Google text to speech (GTTS) library in python
        Upload Sign Image/Images  (SI)
        Save sign image in (I) variable and sign data in (D) variable
        //Search & Retrieval sign data (D) for each sign image (I) from DB
          For II=II: I
                        set SS= sign image
                        For DD=DD :D
                          set ff=sign data
                          If SS=ff then Append sign data to the phrase variable
                        NEXT
          NEXT
        Phrase= sign data for each sign image
        Convert text in phrase variable to speech by GTTS
        Output speech by mic & Recognized Text
```

Figure 4. Pseudo code for translating ArSL into speech.

```
def speech_to_text(request):

# template_name = 'speech_to_text.html'
template_name = 'speech_to_text.html'

data = request.POST.get('record')
import speech_recognition as sr

# get audio from the microphone
r = sr.Recognizer()
with sr.Microphone() as source:
    print("Speak:")
    audio = r.listen(source)
try:
    output = " " + r.recognize_google(audio,language ="ar-AR")
except sr.UnknownValueError:
    output = "Could not understand audio"
except sr.RequestError as e:
    output = "Could not request results; {0}".format(e)
data =output
```

Figure 5. A sample code in Python.

images, with the meaning of each sign image shown next to each on the screen, as shown in Figure 8.

When the "الاشارة الي صوت" link is clicked on the homepage of the application (Figure 6), an interface of the second subsystem appears, as shown in Figure 9. This subsystem translates ArSL into speech and text.

As shown in Figure 9, we click the "اختر صورة" button, and then click the "تحميل" button, to load the image/images of the sign to be translated into speech. For example, we will download the signs for the sentence "أنا احب كرة القدم". After that, we click the "تحويل الي صوت" button and the speech corresponding to these selected signs will be output. In addition, the corresponding text will be displayed, as shown in Figure 10.

In addition, a "مسح الكل" button will appear, to delete all the contents of the page to translate other new signs.

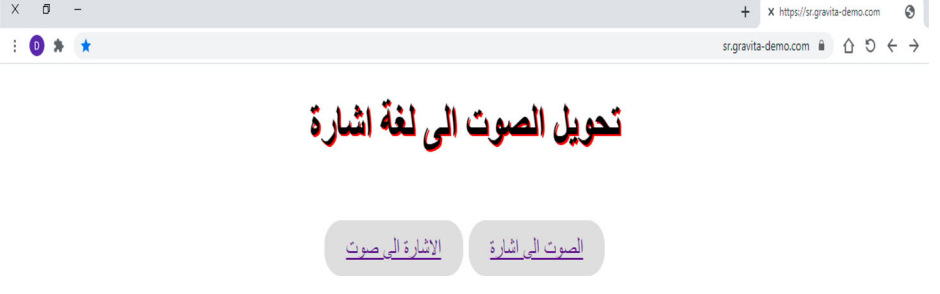


Figure 6. Web application home page.

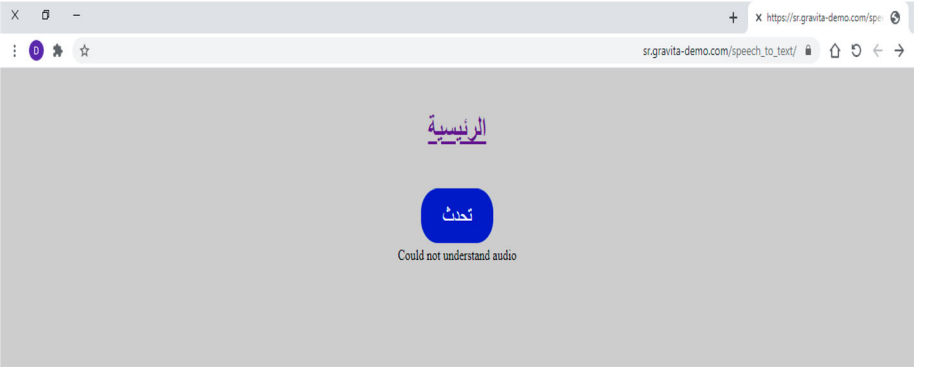


Figure 7. Speech to ArSL subsystem interface.

5. System evaluation

Expert evaluations were widely used to evaluate the outputs of the proposed system, validate it and ensure its applicability by presenting it to a number of ArSL experts, who selected a number of desired inputs and outputs for the purposes of evaluating the system. Evaluation was conducted in two phases: first, they operated the system and input an ordered set of Arabic speech (isolated words, complete sentences) to see if the system will output the equivalent signs. Then, they identified a sign image/sequence of sign images to make sure the system will output the speech matching such inputs. The number of Correct and incorrect outputs were recorded. After that, the recognition rate was calculated using the following equation (Anggraini et al., 2018):

$$RR = \frac{NCorrect}{NTotal} \times 100\%, \tag{1}$$

where RR is the Recognition Rate, Ncorrect is the number of signs/spoken words correctly recognized, and NTotal is the total number of signs/spoken words input to the system.

The findings show that the proposed system was able to translate ArSL with a recognition rate of 99% whereas only 1% of the inputs were not recognized. This high accuracy rate mainly due to two major factors: the accuracy and quality of the speech recognition engine and the accuracy of GTTS engine. Those experts confirmed that the system is reliable and applicable and can be used as a practical and educational tool to recognize ArSL.



Figure 8. Results of translating speech into sign language.

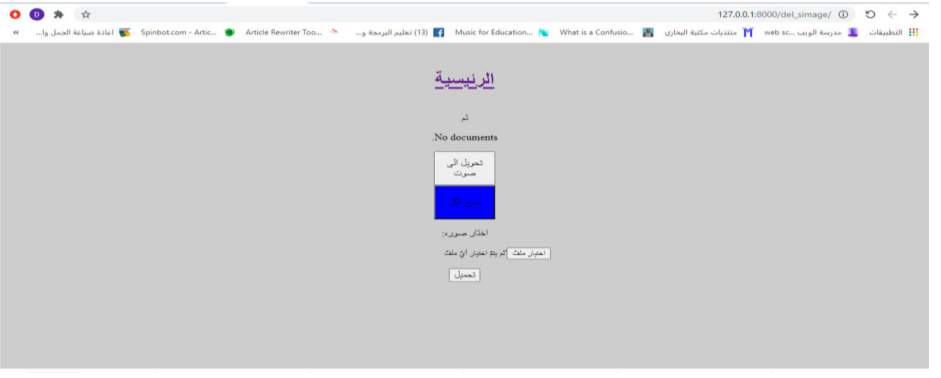


Figure 9. ArSL to speech subsystem interface.

6. Results and discussion

The objective of this study is to develop a web-based ArSL translation system as a contribution to the automatic recognition of ArSL and an attempt to solve the difficulties and challenges faced by deaf students in educational environments in terms of interacting and communicating with normal peoples, and to facilitate their inclusion in the educational community more effectively as well as to encourage and support the learning of ArSL. To achieve that goal, we divided it into sub-goals and phrased them as research questions. Here are the research questions we tried to answer:

Q1: What are the specifications of the proposed ArSL learning system?
Q2: How effective is the proposed system in teaching ArSL to the participants of the study?

In the following section, we will discuss in detail the experimental findings of the study.

6.1 Participants

The participants were a representative random sample consisting of 60 male and female hearing disabled students aging 6–8 years in hearing-impaired School cycle1, Al Mansoura City. Participants were divided into two equal groups, the control group ($n = 30$) and the experimental group ($n = 30$), as well as about 15 volunteers (teachers, specialists for the deaf and speech disabled, and normal persons).



Figure 10. Translating ArSL into speech and text as output.

6.2 Experimental results

After ensuring the system's applicability, it was implemented and used by the participants. But before implementing the system, the researchers prepared an illustrated test to measure the participant's acquisition of the signs included in the system. Test questions were objective and illustrated comprising two types of question: multiple choice questions and matching linked shapes. In its primary form, the test comprised 28 items. Test validity was verified by Arbitrators of instructional technology and special education arbitrators, who recommended some modifications to the phrasing of certain vocabulary items. The reliability of the test was verified using Cronbach's Alpha, after implementing the test to a pilot sample. Reliability factor was 0.81, which shows that the test has an acceptable reliability. Difficulty and discrimination levels of each test items were calculated and three items were deleted, so that, in its final form, the test consisted of 25 items with a maximum score of 25 and the available time for answers is about 25 min. Thus, the test became applicable to the research participants. After preparing the illustrated test and ensuring its validity and reliability, the research experiment was implemented as follows: the pre-test was implemented to the participants. After that, the experimental group studied the proposed system whereas the control group studied the same system content in the traditional way. The implementation took twelve weeks in the first term of 2019/2020 school year. Finally, the post-test was implemented to the participants and data were collected for perform the appropriate statistical processing and then answer the questions of the study. To answer the first research question, the proposed system was developed as described in detail in the above sections. To answer the second research question, the following three research hypotheses were verified:

Hypothesis 1: There are statistically significant differences between mean scores of the pre- and post-test of the experimental group in the illustrated test for the favor of the post-test. To validate Hypothesis 1, the T-test was used. Table 3 shows the significance of the differences between the mean scores of the pre- and post-tests of the experimental group students in the illustrated test.

From the results shown in Table 3, the value of (T) calculated at the comparison of the mean scores of the pre- and post-tests for the students of the experimental group in the illustrated test is (11.096), which is statistically significant at level 0.01 for favor of the post-test. Thus, Hypothesis 1 is validated.

Hypothesis 2: There are statistically significant differences between mean scores of the control and experimental groups in the post-test of the illustrated test for the favor of the experimental group. To validate this Hypothesis, the T-test for both groups was used. Table 4 shows the significant differences between the mean scores of the students of the control and experimental groups in the post illustrated test.

From the results shown in Table 4, the value of (T) calculated at the comparison of the mean scores of the students of the control and experimental groups in the post-test of the illustrated test, is equal to (3.045), which is statistically significant at the level 0.01 for favor of the experimental group, Thus, Hypothesis 2 is validated.

Table 3. Results of the *T*-test for the significant differences between pre- and post-test mean scores for the experimental group in the illustrated test.

Test	*n*	mean	SD	"T" Value	Significance level
Pre	30	11.80	1.690	11.096	0.01
Post	30	40.07	9.599		

Table 4. Results of *T*-test for the significance of differences between the mean scores of the control and experimental groups in the post-test.

Group	*n*	mean	SD	"*T*" value	Significance level
Control	30	20.08	9.265	3.045	0.01
Experimental	30	28.57	9.599		

Table 5. Calculating the effectiveness of the proposed web application in acquiring and learning ArSL.

Group	Instrument	*X*	*Y*	*Z*	Effectiveness
Control	Illustrated test	20.081	12.041	25	0.9416
Experimental		23.570	11.80	25	1.3624

Hypothesis 3: The proposed system is effective in teaching ArSL to the participants.

To validate Hypothesis 3, the following Blake equation was used:

$$\text{Effectivness} = \frac{x-y}{z-y} + \frac{x-y}{z}, \tag{2}$$

where x is the mean student scores in the post-test, y is the mean student scores in the pre-test and z is the final test scores.

From Table 5, it is clear that the effectiveness of the traditional method to teaching ArSL to the control group was 0.9416, which is relatively representative but less than the effectiveness threshold (>1.2), which makes the traditional method lower in terms of acquiring and learning sign language. However, the effectiveness of the proposed system used to teach ArSL to the experimental group was 1.3624, which is greater than the effectiveness threshold (>1.2). This indicates that the proposed system is more effective in acquiring and teaching ArSL. We should note that another method was used to test and evaluate the proposed system by the Deaf students, who interacted with the volunteers during two different evaluation sessions each lasted over 3 h in the presence of an ArSL specialist. During these evaluation sessions, for the purposes of the test, we carefully created several realistic scenarios and situations related to what these students learned, such as requesting different types of information or services or simulating different types of problems that require the interaction of the hearing disabled users with the volunteers. At first, the volunteers looked at signs implemented the hearing disabled students, such as Where is the pen? After then, the volunteers were asked to convert these signs into spoken words. In addition, the deaf students looked at certain signs and were asked to identify them, such as "انعطف يمينًا " (Turn right) and the students did as they were told and so on. After that, certain questions were asked to the participants through the questionnaire prepared by the authors using Google Forms at the following link: https://forms.gle/dWecksaPtoGgCD2c6. Through this questionnaire, several aspects of the system performance can be assessed. The questionnaire was prepared in Standard Arabic and ArSL. The questionnaire contains six simple questions with a scale of five answers (1 = Poor, 2 = Fair, 3 = Good, 4 = Very Good and 5 = Excellent). The questions were designed to identify whether the system is easy-to-use, helpful, has a low response time, gives correct translations and is suitable to learn ArSL. The objective of the last question is to identify whether individuals are ready to use it on a daily basis. Figure 11(a–f) shows the results of the questionnaire.

7. Conclusion

Teaching and learning Sign Language is one of the main challenges faced by the Deaf students, their teachers and normal people interacting with them. So, this paper proposed a two-way system to translate ArSL into speech and translate speech into ArSL. The system aims at teaching ArSL signs to the Deaf and also to normal people, which helps filling the communication gap between them. So, this proposed system may motivate parents, friends, teachers and normal persons to learn Sign Language to communicate with the hearing disabled in the family or in the society in general, and in classrooms in particular. The application supports several platforms. To identify the feasibility of this system in educational environments, tests and questionnaire were implemented to collect data from the Deaf students and their teachers in classrooms. Results show that the proposed system is accurate and effective. Moreover, all system users found it

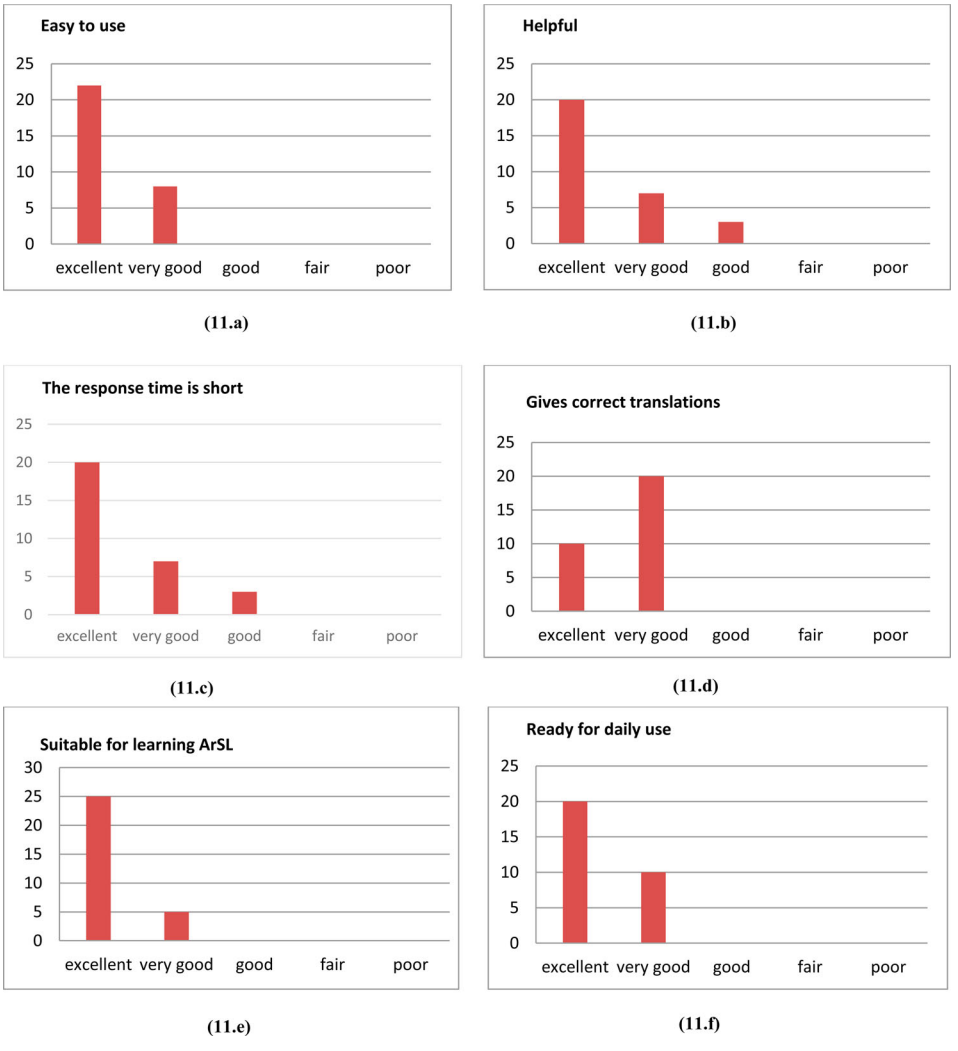


Figure 11. The histograms of the collected data: (a) easy to use, (b) helpful, (c) the response time is short, (d) gives correct translations, (e) suitable for learning ArSL, and (f) ready for daily use.

useful and suitable for learning ArSL. In addition, the system was evaluated by the experts of the field and the results show that the proposed system was able to translate ArSL in both ways with a recognition rate of 99%. We intend to develop the system more in future to meet various teaching and learning scenarios and to provide it as an iPhone app.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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